

View Reviews

Paper ID

2690

Paper Title

SGPA: Spectrogram-Guided Phonetic Alignment for Feasible Shapley Value Explanations in Multimodal Large Language Models

Track Name

Interspeech 2026 Main Track

Reviewer #2

Questions

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Agreement accepted

6. Type of scientific or technological contribution Please assess the type of contribution. Tick all applicable options. If the type of contribution is not listed, or if you are unable to determine it, please select the last option (only).

New model, algorithm, architecture, or recipe

7. Originality (novelty) score

3: Sufficiently novel

8. Technical or methodological correctness score

4: Technically solid

9. Clarity of presentation score

3. Clear enough, could benefit from some revision

10. Overall recommendation

5: Accept: I think this paper should be accepted

11. Feedback and justification of your numerical scores Please provide your comments to the authors here, including justification for your

numerical scores. We recommend you to use the following template (copy & paste it to the text field). -----

----- **Summary of the paper: Major strengths or weaknesses (taking into account originality, technical correctness, clarity): Minor issues: Justification for the overall recommendation:** -----

This paper addresses the computational intractability of applying Shapley Value (SV) attribution to end-to-end audio language models. The authors propose Spectrogram-Guided Phonetic Alignment (SGPA), a four-stage pipeline that maps raw audio to word-aligned segments using CTC forced alignment and spectral boundary refinement. The method aims to reduce dimensionality, eliminate semantic dilution of individual audio frames, and mitigate boundary artifacts.

Reviewer #3

Questions

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6. Type of scientific or technological contribution Please assess the type of contribution. Tick all applicable options. If the type of contribution is not listed, or if you are unable to determine it, please select the last option (only).

New model, algorithm, architecture, or recipe

7. Originality (novelty) score

2: Minor novelty

8. Technical or methodological correctness score

1: Severely flawed

9. Clarity of presentation score

3. Clear enough, could benefit from some revision

10. Overall recommendation

2: Reject: I think this paper should be rejected

11. Feedback and justification of your numerical scores Please provide your comments to the authors here, including justification for your numerical scores. We recommend you to use the following template (copy & paste it to the text field). -----

----- **Summary of the paper: Major strengths or weaknesses (taking into account originality, technical correctness, clarity): Minor issues: Justification for the overall recommendation:** -----

Summary of the paper:

The paper presents a new interpretation technique, which uses word-level input segments instead of timeframes to estimate the Shapley values and identify influential inputs. The proposed solution is evaluated on a subset of the VoiceBench corpus, using the LFM2-Audio-1.5B model.

Major strengths or weaknesses (taking into account originality, technical correctness, clarity):

The main strength of the paper is the demonstrated speed-up achieved by switching to word segments as basic interpretation units, demonstrating that Shapley importance scores of audio-based LLMs can be efficiently estimated in this way.

The main weakness of the paper is the evaluation protocol: the solution is mainly evaluated in terms of computational cost and speed-up and not as an interpretation technique. The authors report that using their technique “systematically alters attribution concentration”, which is an alarming observation, the question arises: which one is correct (if either of them is reliable)? To assess the correctness of the proposed solution, I strongly suggest that the authors employ standard metrics like Infidelity or Sensitivity proposed for evaluating interpretation techniques. Furthermore, since the proposed solution is heavily reliant on the quality of word segmentation, it should also be tested (in my experience, CTC-based segmentations are

often erroneous, due to the blank symbol's dominance). Since the segmentation model is a standalone system using a wav2vec 2.0 model and some additional refinements, it would be important to test it on some standard benchmarks to demonstrate that its segmentation is really high quality. Lastly, the fact that only synthesised audio was used for the experiments questions how well the solution would work on real speech.

Minor issues:

The abstract, in my opinion, is way too technical. It should be redrafted for a more general audience.

In 2.1, I was expecting some information about how pronunciation variants are handled .

In 2.2, the authors state that the wav2vec 2.0 model was chosen for “its robust CTC alignment quality”, but cite no supporting paper (CTC-based word-segmentation is still an open issue, actively researched, and I would not call it robust without ample empirical evidence).

In 2.2, no information is given on how the segments aligned with the Blank label are handled.

In section 2.3, line 127, the authors state that alpha and beta parameters are “empirically tuned”, without giving information on what data and the protocol for evaluating (was a development data set used here or were these tuned on the evaluation data, or some other corpus?)

In Figure 2 it is hard to compare the two figures, and the large empty gap in the top picture reduces readability.

In 3.3.1, the entropy is divided by the square root of n , but since the calculations are in logarithmic space, it has a different meaning. Could the authors provide more information on why they think that using this version of the geometric mean would be the best way to compensate for the length dependence?

Justification for the overall recommendation:

While the proposed method looks interesting, and offers significant speed-up over the standard Shapley interpretation, its correctness remains unclear. Due to the missing evaluation of the segmentation module and the assessment of the final Shapley values' Infidelity or Sensitivity it is unclear if the proposed algorithm is producing correct interpretations. Due to these issues, I recommend the rejection of this article.

Reviewer #4

Questions

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6. Type of scientific or technological contribution Please assess the type of contribution. Tick all applicable options. If the type of contribution is not listed, or if you are unable to determine it, please select the last option (only).

New model, algorithm, architecture, or recipe

New corpus, software toolkit, or experimental protocol

New application

7. Originality (novelty) score

2: Minor novelty

8. Technical or methodological correctness score

3: Minor issues but credible results

9. Clarity of presentation score

3. Clear enough, could benefit from some revision

10. Overall recommendation

3: Weak Reject: I am leaning to reject this paper

11. Feedback and justification of your numerical scores Please provide your comments to the authors here, including justification for your numerical scores. We recommend you to use the following template (copy & paste it to the text field). -----

----- **Summary of the paper: Major strengths or weaknesses (taking into account originality, technical correctness, clarity): Minor issues: Justification for the overall recommendation:** -----

Paper proposes a four-stage pipeline for converting raw audio into word-aligned segments. Experiments using a voice-derived dataset demonstrates efficiency and effect of segmentation.

Strengths:

- 1) Paper addresses explainability in audio-based LLMs.
- 2) Methodology is clear and achievable
- 3) Evaluation shows computational improvements

Weaknesses:

- 1) The approach combines existing techniques
- 2) Only one model and a relatively small, synthetic (TTS-generated) dataset are used.

Minor issues:

- 1) Hyperparameters are empirically chosen without sensitivity analysis.
- 2) Dataset size is relatively small for generalization.
- 3) Minor formatting inconsistencies.

While promising, the work would benefit from broader evaluation and stronger comparative analysis.

Reviewer #5

Questions

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comments will be forwarded to the authors. They will also help
Interspeech 2026 to decide the outcome of the paper, and justify the
decision for the authors. If the paper is accepted, the comments should
guide the authors to improve the presentation of their final manuscript.
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contents confidential. 2. Be constructive and avoid offensive language.
3. Do not try to actively discover the identities or affiliations of the

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Agreement accepted

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New model, algorithm, architecture, or recipe

7. Originality (novelty) score

3: Sufficiently novel

8. Technical or methodological correctness score

2: Significant flaws that may call results into question

9. Clarity of presentation score

2. Difficult to read

10. Overall recommendation

3: Weak Reject: I am leaning to reject this paper

11. Feedback and justification of your numerical scores Please provide your comments to the authors here, including justification for your numerical scores. We recommend you to use the following template (copy & paste it to the text field). -----

----- **Summary of the paper: Major strengths or weaknesses (taking into account originality, technical correctness, clarity): Minor issues: Justification for the overall recommendation:** -----

The paper addresses explainability/interpretability in multimodal LLMs.

Major strengths:

- Addresses an important open question and well worth new contributions.
- The motivation and rationale of the proposed method is legitimate.
- Use of language (on a grammar level) is very clear.

Weaknesses:

- This is LLM research, which on the boundary of what Interspeech covers.
- Insufficient explanation of basic background. In particular, basics of the Shapley value are not explained, so the meaning of "player" and "coalition"

in Section 2 are unclear without looking it up from background literature. Similarly, sections 3 and 4 cannot be understood without extensive background knowledge in Shapley methodology. I'm thus unable to determine the extent to which the results are meaningful, without resorting to background literature. For example, the article does not explain whether or why higher or lower entropy is beneficial (re Fig 3). From the text, I'm deducing that your claim is that more focused attribution of influential parts of the signal is preferred. I was however unable to determine why that would always be the case? Can't there be sentences where all words are equally important, such that enforcing focus on single words would bias the results away from the truth?

- Evaluated only on synthesized voices. It is unclear whether results generalize also to real speech. I would expect synthetic speech to be easier to segment, so in effect, you have chosen an easier target. This is naturally not a central focus of the paper, but still, for real-world applicability real speech would be required.

Minor issues:

- Fig 1 refers to a "Hello world" sentence, but the waveform does not at all resemble such a signal. The font size in the figure is very small. It is claimed that this figure provides an overview for the SGPA pipeline, but it is confusing; there are supposed to be four steps, but there are three parts (a,b,c) in the figure. The connection between the steps and the figure is unclear.

- By "wall-clock time" I suppose you refer to "running time"?

- What is the normalization of the entropy by $\sqrt{n}$ based on? I have seen this format before, but cannot remember what the assumptions are to use this normalization and a citation is missing. For independent observations the normalization should be scaled by n , but for time series I believe $\sqrt{n}$ is right, but I don't remember why. This is of crucial importance, since changing the scaling could reverse the results.

Reviewer #6

Questions

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New model, algorithm, architecture, or recipe

7. Originality (novelty) score

2: Minor novelty

8. Technical or methodological correctness score

4: Technically solid

9. Clarity of presentation score

3. Clear enough, could benefit from some revision

10. Overall recommendation

4: Weak Accept: I am leaning to accept this paper

11. Feedback and justification of your numerical scores Please provide your comments to the authors here, including justification for your numerical scores. We recommend you to use the following template (copy & paste it to the text field). -----

----- **Summary of the paper: Major strengths or weaknesses (taking into account originality, technical correctness, clarity): Minor issues: Justification for the overall recommendation: -----**

Summary of the Paper:

In order to combat the intractability of Shapley Value attribution of audio inputs using native audio tokenization, this paper proposes a four-stage pipeline aiming at reduction of number of audio segments considered in Shapley Value (SV) calculation. By leveraging the transcription of a speech audio, the proposed SGPA method segments speech audio into parts naturally separately by modified word boundary, which is a boundary refined

from the word boundary by looking for the position in audio spectrum that is close to a word boundary and maintains low-energy spectral stableness. Controlled diagnosis was conducted on a subset of one-sentence audio files from VoiceBench dataset with LFM2-Audio-1.5B as the backbone audio foundation model, and experiments showed clear improvement in SV approximation, bringing originally time-consuming operations into regime of feasible computation with practical runtime (43x reduction in computation and 28x in wall-time performance).

Major strengths:

1. An intuitive, yet new and effective approach to reducing number of segments in an audio input considered for Shapley Value attribution approximation, making the SV value approximation tractable for realistic applications.
2. Reduction in audio segments backed by semantic alignment, that is, aligning segments with words expressed by the audios, thereby improving the interpretability of the calculated attribution values.

Major weaknesses:

1. It is unclear to the reviewer whether Stage 3: Spectral Boundary Refinement is actually necessary. The authors make sensible hypothesis on this stage being effective at reducing audible artifacts; however, it is not clear if audible artifacts would lead to abnormality in attribution values, nor is it clear whether such a stage would prevent a shift in the interpretability of the resulting SV scores.
2. Problem 1 is compounded by the absence of a baseline that considers a naive word boundary without spectral refinement, or other possible alternative approaches that reduce number of segments considered in SV approximation.
3. Although segmenting the audio input by spoken words naturally makes the attribution scores interpretable by human, compared to native audio tokenization, it is unclear whether the estimated SV values correlate with human expectation or other more robust measure of the "quality" of the attribution scores. It is possibly out-of-scope of this paper to quantify the evaluation of SV scores for audio input, but without actual examples and metrics, the usefulness of SV scores calculated by SGPA remains unclear.

4. The reduction in runtime is to be expected; however, it is unclear if the results on attribution statistics are due to the specific design of SGPA pipeline, or simply due to a reduction in number of segments considered in the calculation. This compounds the problem stated in Point 2, that is, without competitive baseline for comparison, the value of SGPA pipeline beyond expected reduction in computation time remains obscure.

To be clear, the reviewer doesn't consider problems stated here as diminishing the value of the main contribution from the paper, but would suggest considering a shift in the focus of the analysis from change in SV statistics (expected) to improvement in SV interpretability by humans, thereby increasing the reference value for future researches adopting this method.

Minor issues:

1. Figure 2, bottom plot, "Count of Count...", is this a typo?
2. Please specify the value for the half-width (δ) used in the pipeline.
3. Optionally, please consider clarifying against what output is the Shapley Value calculated. It is reasonable to assume the SV score is calculated based on change in the transcription text, but clarifying the objective in Section 3 in Model or Dataset subsection would make the paper easier to follow.

Justification for Overall Recommendation:

This reviewer would still consider the paper as acceptable for publication given its contribution to a core application of Shapley Value to audio language models, which is to this reviewer's knowledge not extensively studied in published records. However, the weak choice of baseline and the analysis on mostly expected results do limit the potential of the paper as a much more robust reference paper for other researches.